

Digital Logic Assignment Solutions

Assignment 2

October 29, 2025

1 Simplify Boolean functions

1.1 Simplify the following four-variable Boolean functions to simplest POS form using K-map method: $F(A, B, C, D) = \prod(0, 2, 6, 11, 13, 14, 15) + d(1, 3, 9, 10, 12)$.

Get the K-map:

$AB \backslash CD$	00	01	11	10
00	0	X	X	0
01	1	1	1	0
11	X	0	0	0
10	1	X	0	X

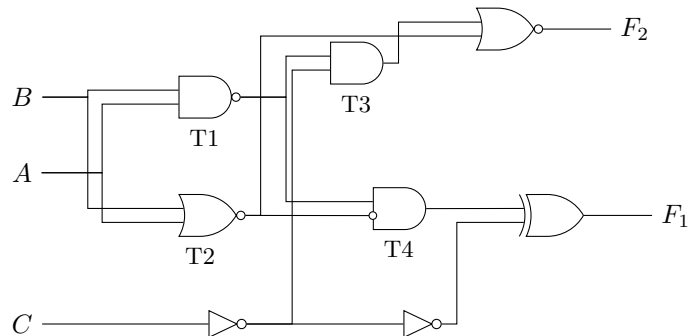
From K-map get the F':

$$F' = A'B' + CD' + AD$$

From F' get the F:

$$\begin{aligned} F &= (F')' \\ &= (A'B' + CD' + AD)' \\ &= (A'B')'(CD')'(AD)' \\ &= (A + B)(C' + D)(A' + D') \end{aligned}$$

2 Obtain the Boolean expressions from Circuit



$$\begin{aligned}
T1 &= (AB)' \\
T2 &= (A + B)' \\
T3 &= TC' \\
T4 &= T1T2'
\end{aligned}$$

2.1 Evaluate the outputs F1 as a function of the 3 inputs.

$$\begin{aligned}
F_1 &= T4 \oplus C \\
&= (AB)'(A + B) \oplus C \\
&= (A' + B')(A + B) \oplus C \\
&= (A'B + AB') \oplus C \\
&= A \oplus B \oplus C
\end{aligned}$$

2.2 Evaluate the outputs F2 as a function of the 3 inputs.

$$\begin{aligned}
F_2 &= (T2 + T3)' \\
&= ((A + B)' + (AB)'C')' \\
&= (A'B' + (A' + B')C)' \\
&= (A'B' + A'C' + B'C')' \\
&= AB + AC + BC \\
&= A(B + C) + BC
\end{aligned}$$

(The answer is not unique)

2.3 List the truth table of the 3 input variables.

A	B	C	T1	T2	T3	T4	F1	F2
0	0	0	1	1	1	0	0	0
0	0	1	1	1	0	0	1	0
0	1	0	1	0	1	1	1	0
0	1	1	1	0	0	1	0	1
1	0	0	1	0	1	1	1	0
1	0	1	1	0	0	1	0	1
1	1	0	0	0	0	0	0	1
1	1	1	0	0	0	0	1	1

- 3 Design a combinational circuit with three inputs, x, y, and z, and three outputs, A, B, and C. When the binary input is 0, 1, or 2, the binary output is two greater than the input. When the binary input is 3, 4, 5, 6, or 7, the binary output is one less than the input.

3.1 List the truth table

X	Y	Z	A	B	C
0	0	0	0	1	0
0	0	1	0	1	1
0	1	0	1	0	0
0	1	1	0	1	0
1	0	0	0	1	1
1	0	1	1	0	0
1	1	0	1	0	1
1	1	1	1	1	0

3.2 K-map simplification for A

$A \backslash BC$	00	01	11	10
0	0	0	0	1
1	0	1	1	1

From K-map get the boolean functions:

$$A = XZ + YZ'$$

3.3 K-map simplification for B

$A \backslash BC$	00	01	11	10
0	1	1	1	0
1	1	0	1	0

From K-map get the boolean functions:

$$B = Y'Z' + YZ + X'Z$$

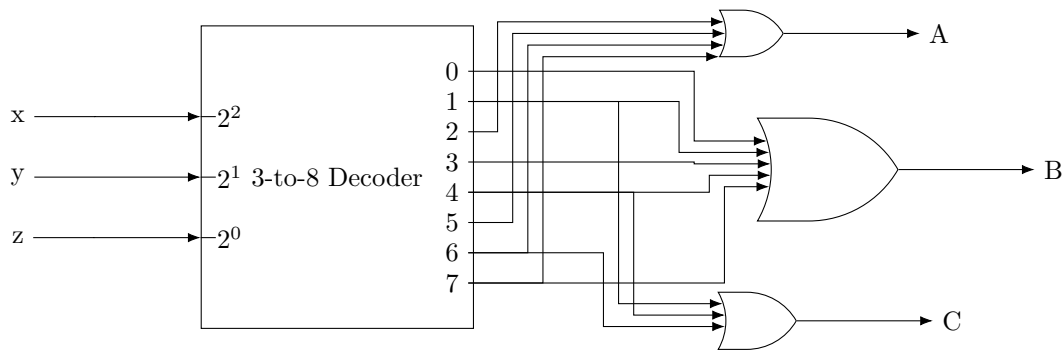
3.4 K-map simplification for C

$A \backslash BC$	00	01	11	10
0	0	1	0	0
1	1	0	0	1

From K-map get the boolean functions:

$$C = XZ' + X'Y'Z$$

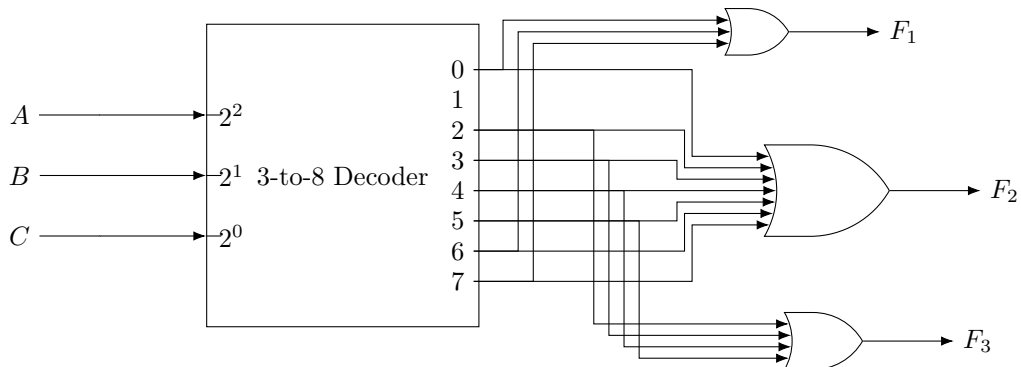
3.5 Draw the logic diagram



- 4 A combinational circuit is defined by the equations $F_1 = AB + A'B'C'$, $F_2 = A + B + C'$, $F_3 = A'B + AB'$. Design a circuit which implements these three equations using a decoder and OR gates external to the decoder, and draw the logic diagram.

4.1 Draw the logic diagram

$$\begin{aligned}
 F_1 &= AB + A'B'C' &= \Sigma m(0, 6, 7) \\
 F_2 &= A + B + C' &= \Sigma m(0, 2, 3, 4, 5, 6, 7) \\
 F_3 &= A'B + AB' &= \Sigma m(2, 3, 4, 5)
 \end{aligned}$$



- 5 An 8:1 multiplexer has inputs A, B, and C connected to the selection inputs S2, S1, and S0, respectively. The data inputs I0 through I7 are as follows: I1=I2=0; I3=I5=I7=1; I0=I4=D; and I6=D'. Determine the Boolean function F(A, B, C, D) that the multiplexer implements.

5.1 Write down the truth table

A	B	C	D	F(A,B,C,D)
0	0	0	0	0
0	0	0	1	1
0	0	1	0	0
0	0	1	1	0
0	1	0	0	0
0	1	0	1	0
0	1	1	0	1
0	1	1	1	1
1	0	0	0	0
1	0	0	1	1
1	0	1	0	1
1	0	1	1	1
1	1	0	0	1
1	1	0	1	0
1	1	1	0	1
1	1	1	1	1

5.2 Simplify the Boolean function with K-map

AB\CD	00	01	11	10
00	0	1	0	0
01	0	0	1	1
11	1	0	1	1
10	0	1	1	1

From K-map get the boolean functions:

$$F = AC + BC + ABD' + B'C'D$$

5.3 Implement the following Boolean function with only one 4x1 multiplexer and external gates.

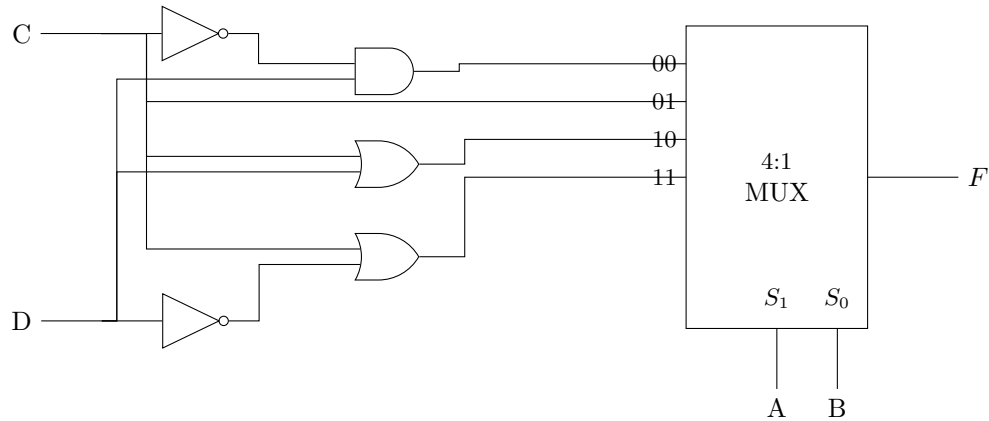
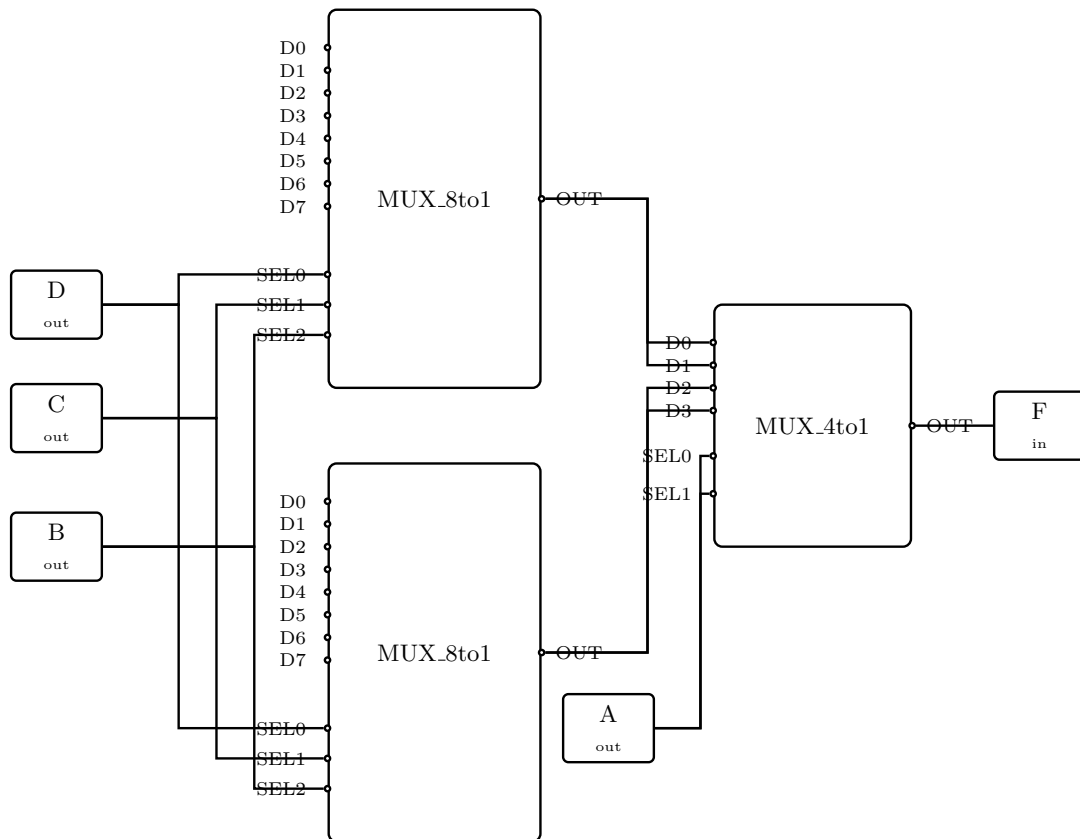


Table 1: 4:1 MUX

0	0	I_0	$\overline{C} D$
0	1	I_1	$C + D$
1	0	I_2	$C + \overline{D}$
1	1	I_3	$\overline{C} \overline{D}$

6 Construct a 16x1 multiplexer with two 8x1 and one 4x1 multiplexers



(The answer is not unique)

- 7 Draw the circuit specified by the following HDL description, use at least one standard standard block if possible.

